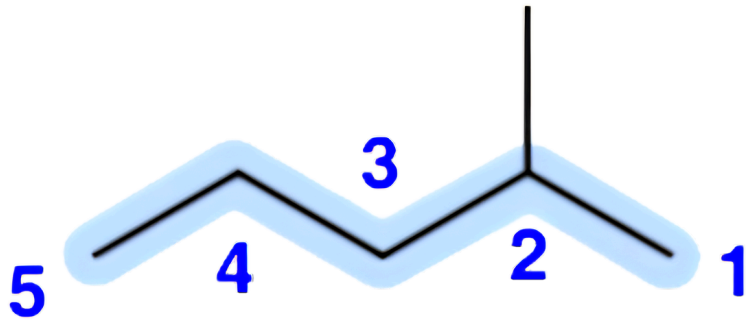
CARD A01 | Structure → Name

Level: Foundation

Molecular Formula: C_6H_{14}

01 NUMBERED PARENT



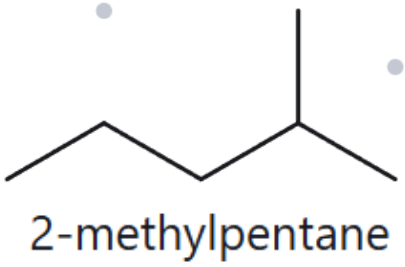
02 NAMING TRAIL

Parent: longest chain = 5 C → pentane.
Numbering: right → left gives 2; left → right gives 4. 2 < 4, so number from the right.
Substituents: methyl on C2.

03 MY NAME

2-methylpentane

04 THE ALKANE CHECK



Returned Name: 2-methylpentane

Verdict: Match ✓

Verification Strip

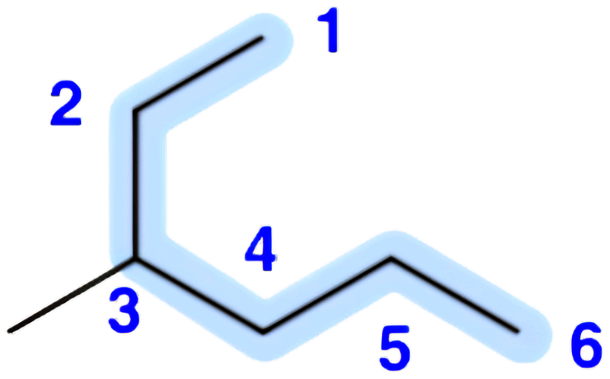
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A02 | Structure → Name

Level: Foundation

Molecular Formula: C_7H_{16}

01 NUMBERED PARENT



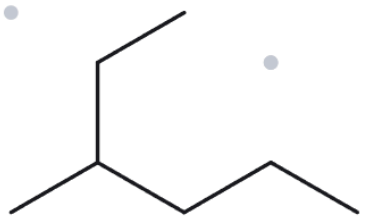
02 NAMING TRAIL

Parent: longest chain = 6 C → hexane. The horizontal path is only 5 C. The longest chain runs through the drawn “ethyl” branch.
Numbering: upper-branch end → right gives 3; right → upper-branch end gives 4. 3 < 4, so number from the upper-branch end.
Substituents: methyl on C3.

03 MY NAME

3-methylhexane

04 THE ALKANE CHECK



Returned Name: 3-methylhexane

Verdict: Match ✓

Verification Strip

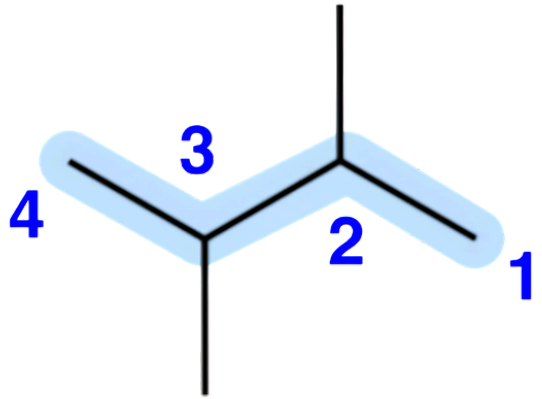
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A03 | Structure → Name

Level: Foundation

Molecular Formula: C_6H_{14}

01 NUMBERED PARENT



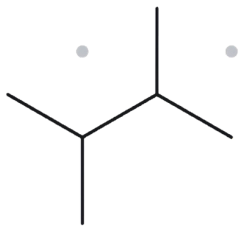
02 NAMING TRAIL

Parent: longest chain = 4 C → butane. No path is longer, because each central carbon carries two methyls.
Numbering: either direction gives 2,3, so the sets tie and either end may start.
Substituents: methyl on C2 and C3 → 2,3-dimethyl.

03 MY NAME

2,3-dimethylbutane

04 THE ALKANE CHECK



Returned Name: 2,3-dimethylbutane

Verdict: Match ✓

Verification Strip

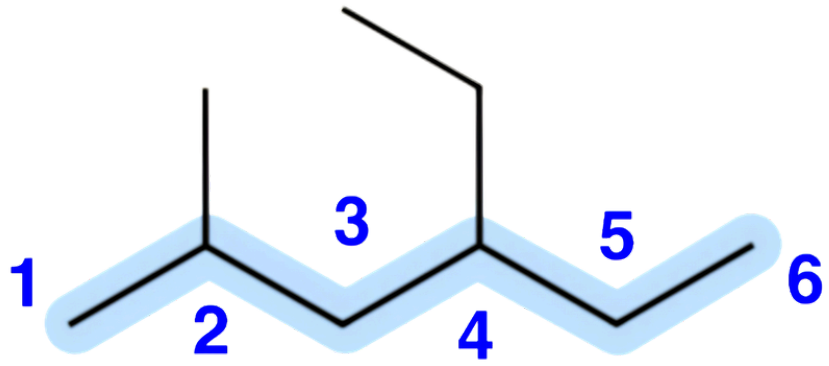
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A04 | Structure → Name

Level: Developing

Molecular Formula: C_9H_{20}

01 NUMBERED PARENT



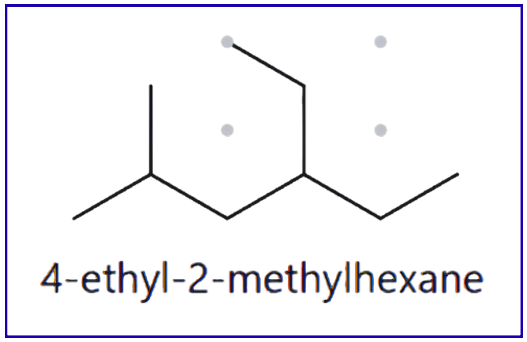
02 NAMING TRAIL

Parent: longest chain = 6 C → hexane. Several 6-C paths exist through the branches, but all carry one methyl and one ethyl, so they give the same name.
Numbering: left → right gives 2,4; right → left gives 3,5. 2 < 3, so number from the left.
Substituents (alphabetical): ethyl on C4, methyl on C2.

03 MY NAME

4-ethyl-2-methylhexane

04 THE ALKANE CHECK



Returned Name: 4-ethyl-2-methylhexane

Verdict: Match ✓

Verification Strip

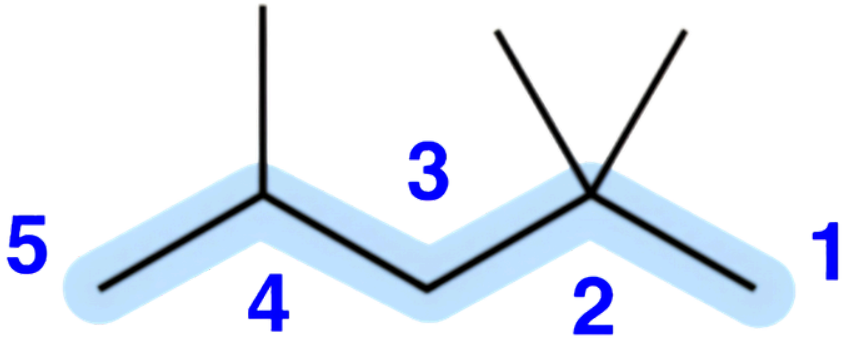
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A05 | Structure → Name

Level: Developing

Molecular Formula: C_8H_{18}

01 NUMBERED PARENT



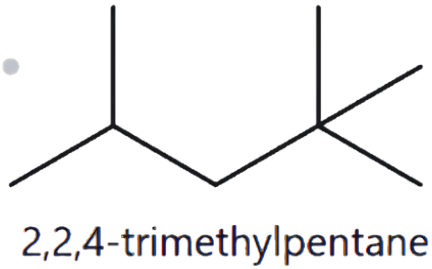
02 NAMING TRAIL

Parent: longest chain = 5 C → pentane.
Numbering: right → left gives 2,2,4; left → right gives 2,4,4. The first term ties (2), and at the second term 2 < 4, so number from the right.
Substituents: methyl on C2, C2 and C4 → 2,2,4-trimethyl.

03 MY NAME

2,2,4-trimethylpentane

04 THE ALKANE CHECK



Returned Name: 2,2,4-trimethylpentane

Verdict: Match ✓

Verification Strip

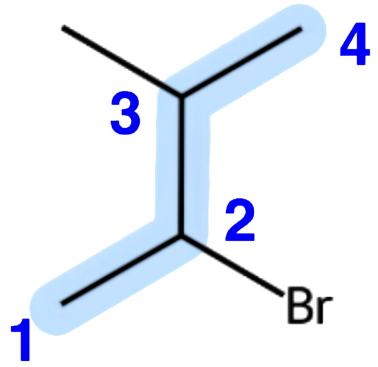
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A06 | Structure → Name

Level: Developing

Molecular Formula: $C_5H_{11}Br$

01 NUMBERED PARENT



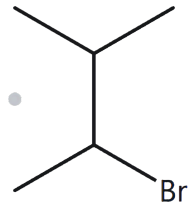
02 NAMING TRAIL

Parent: longest chain = 4 C → butane.
Numbering: both directions give the locant set {2,3}, so the sets tie.
Tie-break: bromo precedes methyl alphabetically, so bromo gets the lower locant (C2).
Substituents (alphabetical): bromo on C2, methyl on C3.

03 MY NAME

2-bromo-3-methylbutane

04 THE ALKANE CHECK



2-bromo-3-methylbutane

Returned Name: 2-bromo-3-methylbutane

Verdict: Match ✓

Verification Strip

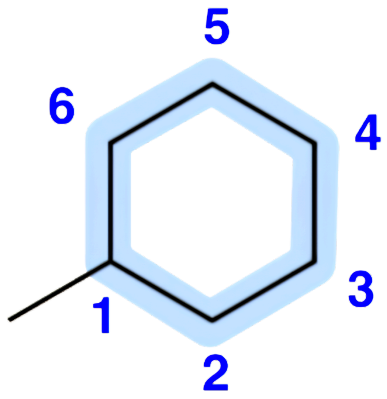
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A07 | Structure → Name

Level: Developing

Molecular Formula: C_7H_{14}

01 NUMBERED PARENT



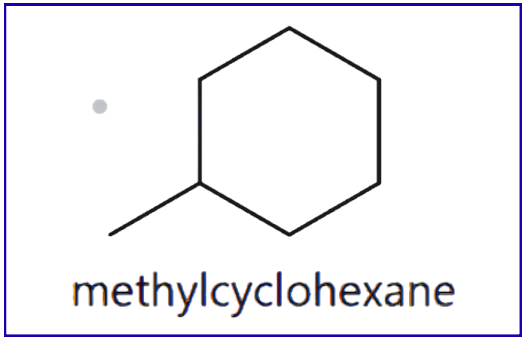
02 NAMING TRAIL

Parent: ring present → ring is the parent; 6 C → cyclohexane.
Numbering: only one substituent, so it takes C1 and no locant is needed (direction is irrelevant).
Substituents (alphabetical): methyl.

03 MY NAME

methylcyclohexane

04 THE ALKANE CHECK



Returned Name: methylcyclohexane

Verdict: Match ✓

Verification Strip

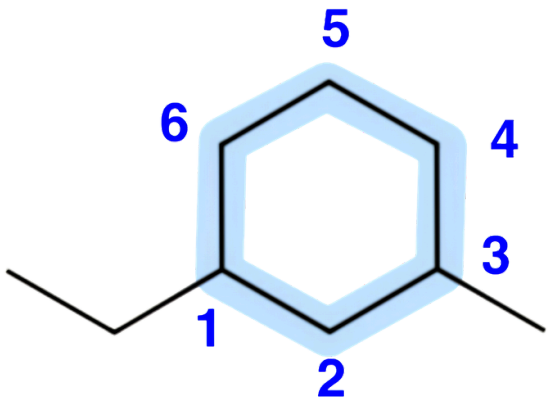
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A08 | Structure → Name

Level: Challenge

Molecular Formula: C_9H_{18}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: ring → cyclohexane (6 C).

Numbering: starting at ethyl (short way) gives {1,3}; starting at methyl (short way) also gives {1,3}, so the sets tie.

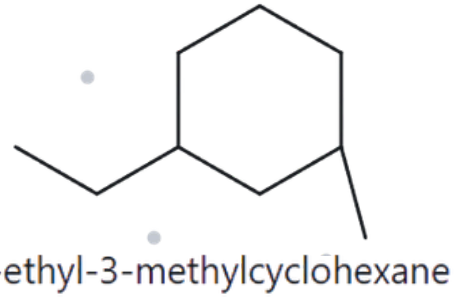
Tie-break: ethyl precedes methyl alphabetically, so ethyl gets C1.

Substituents (alphabetical): ethyl on C1, methyl on C3.

03 MY NAME

1-ethyl-3-methylcyclohexane

04 THE ALKANE CHECK



Returned Name: 1-ethyl-3-methylcyclohexane

Verdict: Match ✓

Verification Strip

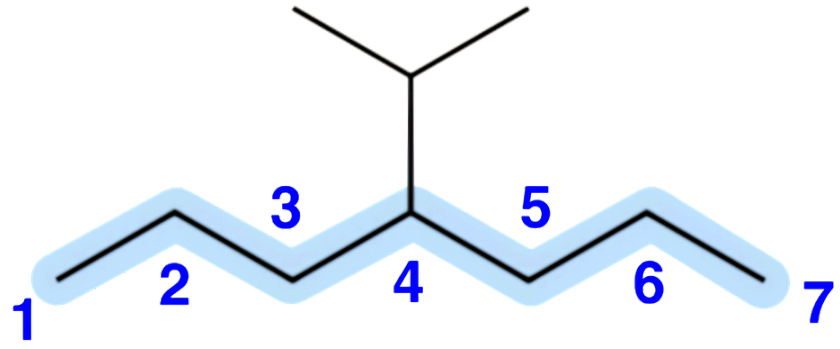
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A09 | Structure → Name

Level: Challenge

Molecular Formula: $C_{10}H_{22}$

01 NUMBERED PARENT



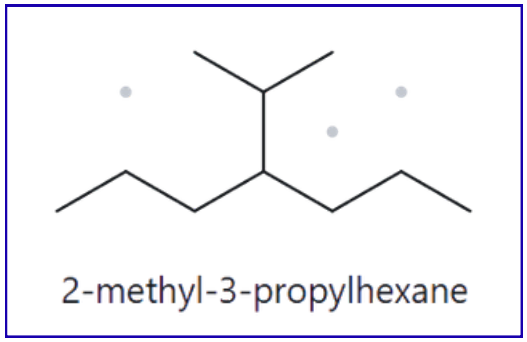
02 NAMING TRAIL

Parent: longest chain = 7 C → heptane. A path through the branch gives only 6 C, so the branch is a substituent.
Numbering: either direction gives 4, so the sets tie.
Substituents: isopropyl [(1-methylethyl)] on C4.

03 MY NAME

4-isopropylheptane

04 THE ALKANE CHECK



Returned Name: 2-methyl-3-propylhexane

Verdict: Mismatch ✗

Correct name: 4-isopropylheptane [4-(1-methylethyl)heptane]. The parent must be the longest chain: 7 carbons (heptane), not 6 (hexane).

Verification Strip

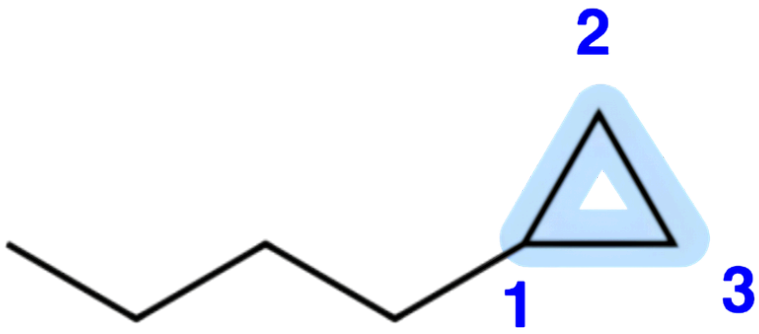
- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A10 | Structure → Name

Level: Challenge

Molecular Formula: C_7H_{14}

01 NUMBERED PARENT



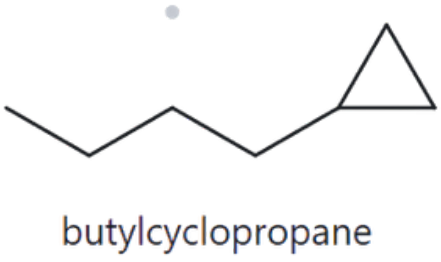
02 NAMING TRAIL

Parent: ring present → ring is the parent; 3 C → cyclopropane. The chain has more carbons (4 C), but the ring takes priority.
Numbering: only one substituent, so it takes C1 and no locant is needed.
Substituents: butyl (4 C).

03 MY NAME

butylcyclopropane

04 THE ALKANE CHECK



Returned Name: butylcyclopropane

Verdict: Match ✓

Verification Strip

- ☑ longest chain/ring
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD B01 | Name → Structure

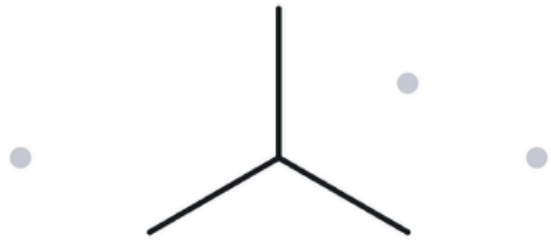
Level: Foundation

Given Name: **2-methylpropane**

01 NAME DECODED

Parent: propane → 3-C chain.
Substituent: methyl on C2.
Expected formula: 3 + 1 = 4 C;
open chain → C_4H_{10}.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_4H_{10} (C: 3 + 1 = 4).
Open chain: C_nH_{2n+2} = C_4H_{10} ✓

04 BACK-CHECK

Returned Name:

2-methylpropane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

CARD B02 | Name → Structure

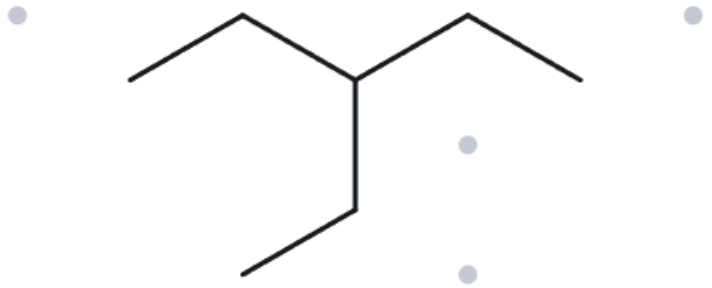
Level: Foundation

Given Name: **3-ethylpentane**

01 NAME DECODED

Parent: pentane → 5-C chain.
Substituent: ethyl on C3.
Expected formula: 5 + 2 = 7 C;
open chain → C_7H_{16}.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_7H_{16} (C: 5 + 2 = 7).
Open chain: C_nH_{2n+2},
n = 7 → 2(7) + 2 = 16 → C_7H_{16} ✓

04 BACK-CHECK

Returned Name:

3-ethylpentane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

CARD B03 | Name → Structure

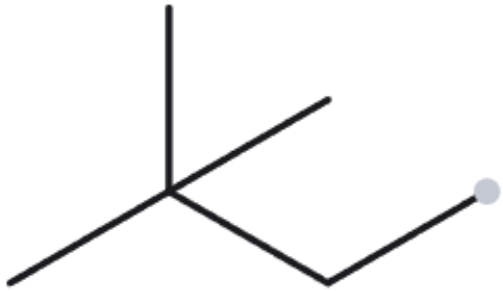
Level: Foundation

Given Name: **2,2-dimethylbutane**

01 NAME DECODED

Parent: butane → 4-C chain.
Substituents: two methyls, both on C2.
Expected formula: 4 + 2 = 6 C;
open chain → C_6H_{14}.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_6H_{14} (C: 4 + 2 = 6).
Open chain: C_nH_{2n+2} = C_6H_{14} ✓

04 BACK-CHECK

Returned Name:

2,2-dimethylbutane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

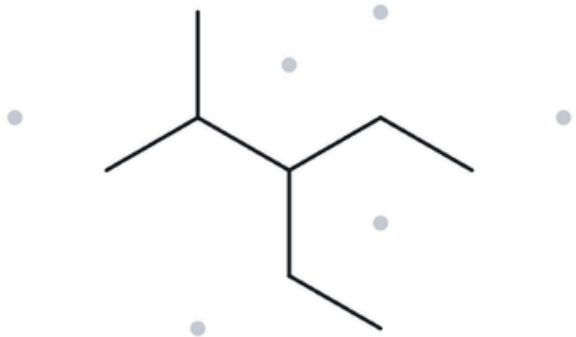
CARD B04 | Name → Structure

Level: Developing | Given Name: **3-ethyl-2-methylpentane**

01 NAME DECODED

Parent: pentane → 5-C chain.
Substituents: methyl on C2; ethyl on C3 (cited alphabetically: ethyl, then methyl).
Expected formula: 5 + 1 + 2 = 8 C;
open chain → C_8H_{18}.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_8H_{18}
(C: 5 + 1 + 2 = 8).
Open chain: C_nH_{2n+2} = C_8H_{18} ✓

04 BACK-CHECK

Returned Name:

3-ethyl-2-methylpentane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

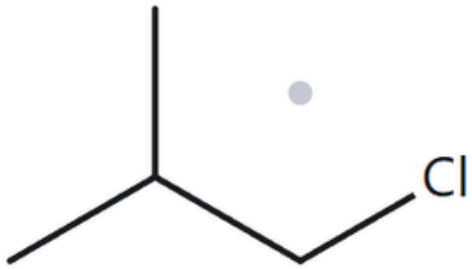
CARD B05 | Name → Structure

Level: Developing | Given Name: **1-chloro-2-methylpropane**

01 NAME DECODED

Parent: propane → 3-C chain.
Substituents: chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 C;
open chain C_4H_{10}; Cl replaces one H → C_4H_9Cl.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_4H_9Cl.
Open chain: C_nH_{2n+2} = C_4H_{10};
minus one H for Cl = C_4H_9Cl ✓

04 BACK-CHECK

Returned Name:

1-chloro-2-methylpropane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

CARD B06 | Name → Structure

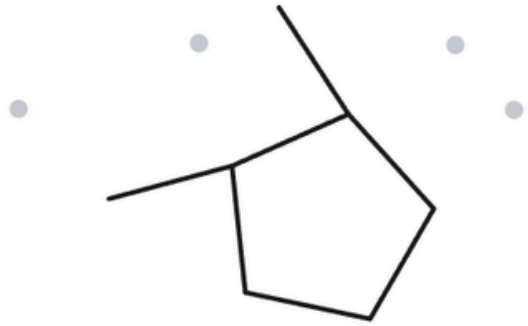
Level: Developing

Given Name: **1,2-dimethylcyclopentane**

01 NAME DECODED

Parent: cyclopentane → 5-C ring.
Substituents: methyl on C1 and C2 (adjacent ring carbons).
Expected formula: 5 + 2 = 7 C;
one ring → C_7H_{14}.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_7H_{14} (C: 5 + 2 = 7).
One ring: C_nH_{2n} = C_7H_{14} ✓

04 BACK-CHECK

Returned Name:

1,2-dimethylcyclopentane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

CARD B07 | Name → Structure

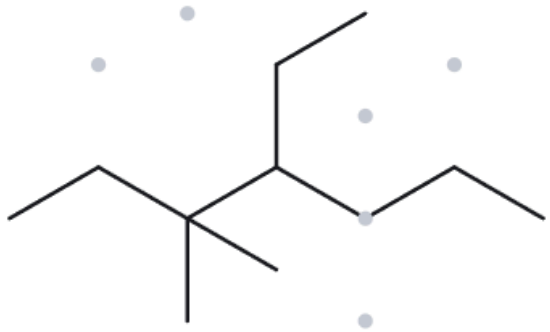
Level: Developing

Given Name: **4-ethyl-3,3-dimethylheptane**

01 NAME DECODED

Parent: heptane → 7-C chain.
Substituents: two methyls on C3 (3,3-dimethyl); ethyl on C4 (cited first alphabetically).
Expected formula: 7 + 2 + 2 = 11 C;
open chain → $C_{11}H_{24}$.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: $C_{11}H_{24}$
(C: 7 + 2 + 2 = 11).
Open chain: C_nH_{2n+2},
n = 11 → 2(11) + 2 = 24 → $C_{11}H_{24}$ ✓

04 BACK-CHECK

Returned Name:

4-ethyl-3,3-dimethylheptane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

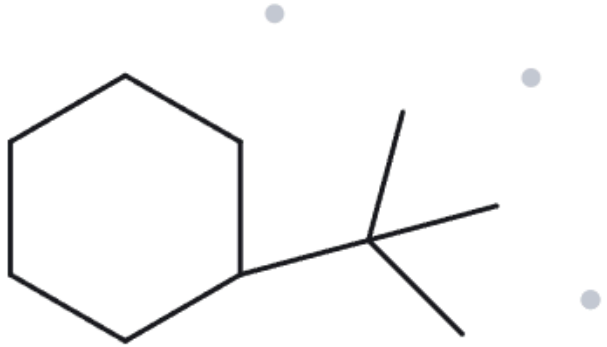
CARD B08 | Name → Structure

Level: Challenge | Given Name: **tert-butylcyclohexane**

01 NAME DECODED

Parent: cyclohexane → 6-C ring.
Substituent: tert-butyl, $-C(CH_3)_3$, a 4-C group attached through its central carbon. It is the only substituent, so no locant is needed.
Expected formula: 6 + 4 = 10 C;
one ring → $C_{10}H_{20}$.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: $C_{10}H_{20}$ (C: 6 + 4 = 10).
One ring: $C_nH_{2n} = C_{10}H_{20}$ ✓

04 BACK-CHECK

Returned Name:

(1,1-dimethylethyl)cyclohexane

Verdict: same compound ✓
written with a different accepted substituent name. tert-Butyl = (1,1-dimethylethyl) = 2-methylpropan-2-yl (Appendix B.3).

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

CARD B09 | Name → Structure

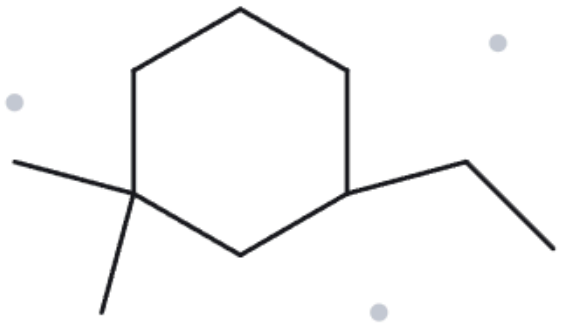
Level: Challenge

Given Name: **3-ethyl-1,1-dimethylcyclohexane**

01 NAME DECODED

Parent: cyclohexane → 6-C ring.
Substituents: two methyls on C1 (1,1-dimethyl); ethyl on C3.
Numbering check: Dimethyl carbon gives {1,1,3}; ethyl carbon gives {1,3,3}. Since {1,1,3} is lower at the first point of difference, the numbering is correct.
Expected formula: 6 + 2 + 2 = 10 C;
one ring → $C_{10}H_{20}$.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: $C_{10}H_{20}$
(C: 6 + 2 + 2 = 10).
One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓

04 BACK-CHECK

Returned Name:

3-ethyl-1,1-dimethylcyclohexane

Verdict: same as the given name ✓

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded

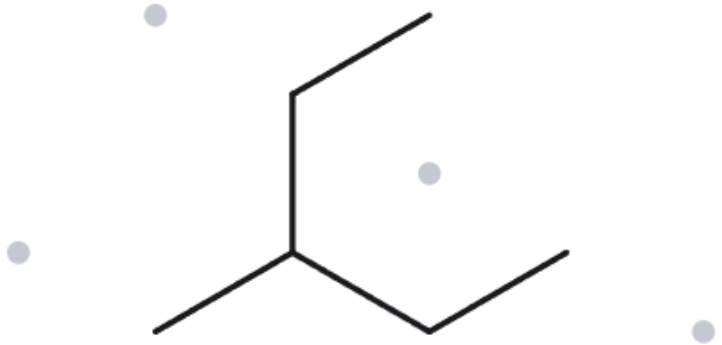
CARD B10 | Name → Structure

Level: Challenge | Given Name: **2-ethylbutane** (deliberately incorrect)

01 NAME DECODED

Parent: butane → 4-C chain.
Substituent: ethyl on C2.
Expected formula: 4 + 2 = 6 C;
open chain → C_6H_{14}.

02 BOND-LINE STRUCTURE (The Alkane)



03 FORMULA CHECK

Counted from drawing: C_6H_{14} (C: 4 + 2 = 6).
Open chain: C_nH_{2n+2} = C_6H_{14} ✓

04 BACK-CHECK

Returned Name:

3-methylpentane

Verdict: differs from the given name ✗
Given name: 2-ethylbutane.
Correct name: 3-methylpentane.
Rule broken: the parent must be the longest continuous chain. The ethyl branch extends the chain through C2, giving 5 C (pentane), so the original C1 becomes a methyl on C3 of pentane.

Verification Strip

- ☑ parent length matches
- ☑ substituent positions match
- ☑ formula agrees
- ☑ back-check recorded